

Apply Spatial Reasoning in Design, Mapping and Layout

Step 10

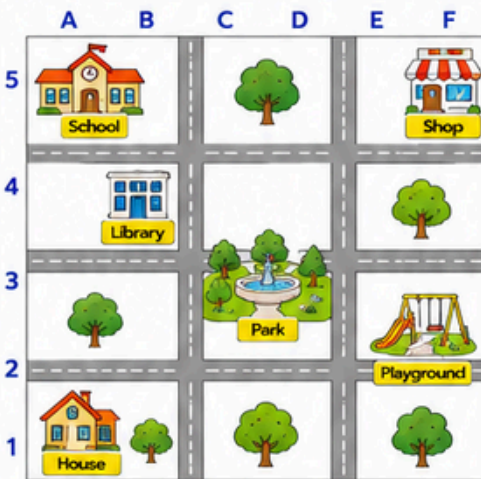
Upper Elementary



Today I am learning to apply spatial reasoning in design, mapping and layout.

1. Read the Map

Look at the map. Use the grid to answer the questions.



- Which building is north of the park?
Answer: _____
- What is east of the library?
Answer: _____
- What is south of the school?
Answer: _____
- Give the coordinates of the playground.
Answer: (____, ____)

2. Follow the Route

a) Start at the school. Follow the directions.

- 3 squares right
- 2 squares up
- 1 square left

Where do you finish?

Answer: _____



b) Create your own route!

Write your directions in order.

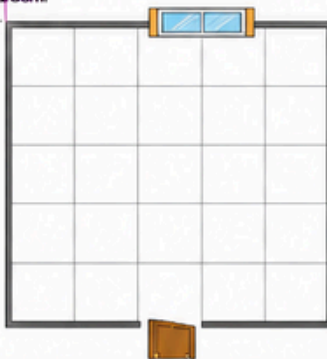
Where did you end?

-
-
-
-

3. Design a Bedroom Layout

Use the grid to design a bedroom.
Place the items in the room.

- Bed
- Desk
- Bookshelf
- Rug
- Door
- Window



Rules:

- ★ Bed must touch a wall.
- ★ Desk must be near a window.
- ★ Bookshelf must not block the door.

4. Floor Plan Challenge

Look at the floor plan below. Answer the questions.



- Which room is closest to the kitchen?
Answer: _____
- Which room is furthest from the entrance?
Answer: _____
- Suggest one improvement to the layout.

5. Scale and Layout

The playground plan is drawn to scale.
Scale: 1 square = 2 metres



- How wide is the playground?
_____ metres
- How long is the playground?
_____ metres
- What is the total area of the playground?
_____ m²

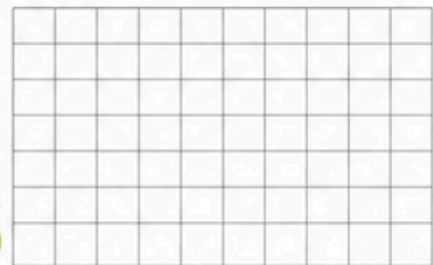
(Remember: Area = length × width)

6. Design a Mini Park

Plan your own mini park on the grid.

Include and label:

- ☐ Playground
- ☐ Trees
- ☐ Pond
- ☐ Walking Path



7. Mapping and Directions

Look at the map. Use compass directions to complete the sentences.



- The shop is _____ of the school.
- The playground is _____ of the library.
- The pond is _____ of the car park.

Words to use: north, south, east, west

8. Think and Explain

- Why are maps useful in everyday life?

- Why do architects and interior designers use plans and layouts?

- How does spatial reasoning help you solve problems in real life?



I can...

- ☐ read and use maps
- ☐ follow and give directions
- ☐ design and plan layouts
- ☐ use scale to measure
- ☐ apply spatial reasoning



Great work!
You are a spatial thinker!



Remember!

Slide (Translation)

Moves in a straight line.



Flip (Reflection)

Turns over a mirror line.



Turn (Rotation)

Turns around a point.



Map Symbols Help Us!



Tree



Water



Building



Road